Дисципліна «Побудова математичних моделей у природознавстві». ЛАБОРАТОРНА РОБОТА № 1. Дослідження математичного апарату і програмних засобів моделювання складних процесів з використанням рівняння математичної фізики. Спеціальність: 113 – «Прикладна математика». Навчально-науковий Фізико-технічний інститут. Кафедра математичного моделювання та аналізу даних

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Завдання 1. Граничні задач для звичайних диференціальних рівнянь другого порядку

Завдання 1.1. Граничні умови 1-го роду (Діріхле)

1) $y'' + xy' - y\sin x = x^2, x \in [0,1], y(0) = 0, y(1) = 1$
2) $y'' - e^x y' + y\cos x = x, x \in [0,\pi], y(0) = 1, y(\pi) = 0$
3) $y'' + y'^{\ln(1+x)} - xy = e^x, x \in [0,2], y(0) = 0, y(2) = 1$
4) $y'' + x^2 y' + y = \sin x, x \in [0,1], y(0) = 1, y(1) = 0$
5) $y'' + \sqrt{x}y' - x^2 y = x^3, x \in [0,1], y(0) = 0, y(1) = 1$
6) $y'' - y' tg\, x + xy = \cos x, x \in \left[0, \frac{\pi}{4}\right], y(0) = 0, y\left(\frac{\pi}{4}\right) = 1$
7) $y'' + e^{-x}y' + xy = x^2, x \in [0,1], y(0) = 1, y(1) = 0$
8) $y'' + xy' - y\ln(1+x) = x, x \in [0,1], y(0) = 0, y(1) = 1$
9) $y'' + y'\sin x + y\cos x = e^x, x \in [0,\pi], y(0) = 1, y(\pi) = 0$
10) $y'' - x^2 y' + \sqrt{x}y = x^3, x \in [0,1], y(0) = 0, y(1) = 1$
11) $y'' + y'^{\ln(x+1)} - x^2 y = \sin x, x \in [0,2], y(0) = 1, y(2) = 0$
12) $y'' + e^x y' + xy = x^2, x \in [0,1], y(0) = 0, y(1) = 1$
13) $y'' + x^3 y' - y = x^3 + 1, x \in [0,1], y(0) = 0, y(1) = 1$
14) $y'' + \sqrt{1+x}y' + x^2 y = \cos x, x \in [0,1], y(0) = 0, y(1) = 1$
15) $y'' - xy' + e^x y = x + 1, x \in [0,1], y(0) = 1, y(1) = 0$
16) $y'' + x^2 y' - xy = \sin x, x \in [0,1], y(0) = 0, y(1) = 1$

Завдання 1.2. Граничні умови 2-го роду (Неймана)

1) $y'' + xy' - y\sin x = x^2, x \in [0;1], y'(0) = 0, y'(1) = 1$

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2) $y'' - e^x y' + y\cos x = x, x \in [0;\pi], y'(0) = 1, y'(\pi) = 0$
3) $y'' + y'\ln(1+x) - xy = e^x, x \in [0;2], y'(0) = 0, y'(2) = 1$
4) $y'' + x^2 y' + y = \sin x, x \in [0;1], y'(0) = 1, y'(1) = 0$
5) $y'' + \sqrt{x}y' - x^2 y = x^3, x \in [0;1], y'(0) = 0, y'(1) = 1$
6) $y'' - y' tgx + xy = \cos x, x \in \left[0;\frac{\pi}{4}\right], y'(0) = 0, y'\left(\frac{\pi}{4}\right) = 1$
7) $y'' + y' e^{-x} + xy = x^2, x \in [0;1], y'(0) = 1, y'(1) = 0$
8) $y'' + y'x - y\ln(1+x) = x, x \in [0;1], y'(0) = 0, y'(1) = 1$
9) $y'' + y'\sin x + y\cos x = e^x, x \in [0,\pi], y'(0) = 1, y'(\pi) = 0$
10) $y'' - x^2 y' + \sqrt{x}y = x^3, x \in [0;1], y'(0) = 0, y'(1) = 1$
11) $y'' + y'\ln(x+1) - x^2 y = \sin x, x \in [0;2], y'(0) = 1, y'(2) = 0$
12) $y'' + e^x y' + xy = x^2, x \in [0;1], y'(0) = 0, y'(1) = 1$
13) $y'' + x^3 y' - y = x^3 + 1, x \in [0;1], y'(0) = 0, y'(1) = 1$
14) $y'' + \sqrt{1+x}y' + x^2 y = \cos x, x \in [0;1], y'(0) = 1, y'(1) = 0$
15) $y'' - xy' + e^x y = x + 1, x \in [0;1], y'(0) = 0, y'(1) = 1$
16) $y'' + x^2 y' - xy = \sin x, x \in [0;1], y'(0) = 1, y'(1) = 0$

Завдання 1.3. Граничні умови 3-го роду (Робіна)

1) $y'' + xy' - y\sin x = x^2, x \in [0;1], y(0) + y'(0) = 0,\ 2y(1) + y'(1) = 1$
2) $y'' - e^x y' + y\cos x = x, x \in [0;\pi], y(0) + 2y'(0) = 1,\ y(\pi) + y'(\pi) = 0$
3) $y'' + y'\ln(1+x) - xy = e^x, x \in [0;2], y(0) + y'(0) = 0,\ y(2) + 3y'(2) = 1$
4) $y'' + x^2 y' + y = \sin x, x \in [0;1], y(0) + y'(0) = 0,\ 2y(1) + y'(1) = 1$
5) $y'' + \sqrt{x}y' - x^2 y = x^3, x \in [0;1], 3y(0) + y'(0) = 1,\ y(1) + y'(1) = 0$
6) $y'' - y' tg\, x + xy = \cos x, x \in \left[0,\frac{\pi}{4}\right], y(0) + 2y'(0) = 0,\ y\left(\frac{\pi}{4}\right) + y'\left(\frac{\pi}{4}\right) = 1$
7) $y'' + e^{-x}y' + xy = x^2, x \in [0;1], 2y(0) + y'(0) = 0,\ 3y(1) + 2y'(1) = 1$
8) $y'' + y'x - y\ln(1+x) = x, x \in [0;1], y(0) + y'(0) = 0,\ y(1) + 2y'(1) = 1$
9) $y'' + y'\sin x + y\cos x = e^x, x \in [0,\pi], y(0) + y'(0) = 0,\ 2y(\pi) + y'(\pi) = 1$
10) $y'' - x^2 y' + \sqrt{x}y = x^3, x \in [0;1], y(0) + y'(0) = 1,\ 2y(1) + y'(1) = 2$
11) $y'' + y'\ln x - x^2 y = \sin x, x \in [0;2], y(0) + y'(0) = 0,\ y(2) + y'(2) = 0$
12) $y'' + e^x y' + xy = x^2, x \in [0;1], y(0) + y'(0) = 0,\ 2y(1) + y'(1) = 1$
13) $y'' + x^3 y' - y = x^3 + 1, x \in [0;1], y(0) + y'(0) = 1,\ y(1) + y'(1) = 1$
14) $y'' + xy' + x^2 y = \cos x, x \in [0;1], 3y(0) + y'(0) = 0,\ y(1) + y'(1) = 1$
15) $y'' - xy' + e^x y = x + 1, x \in [0;1], y(0) + y'(0) = 0,\ y(1) + y'(1) = 1$
16) $y'' + x^2 y' - xy = \sin x, x \in [0;1], y(0) + y'(0) = 1,\ 2y(1) + y'(1) = 0$

Завдання 2. **Граничні задачі для диференціальних рівнянь другого порядку у частинних похідних в декартовій системі координат**

Загальна форма рівняння

$$\frac{\partial}{\partial x}\left(\frac{k_x(x,y)\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{k_y(x,y)\partial u}{\partial x}\right)-q(x,y)u=f(x,y)$$
$$(x,y)\in[A;B]\times[C;D]$$

Задача 2.1.

$$[A;B]\times[C;D]=[1,2]\times[1;2],$$
$$k_x=1+x+\frac{y}{2},\ \ k_y=1+xy+\frac{1}{2},\ \ q=-(1+x^2+y^2),\ \ f=\sin(xy),$$
$$u(1,y)=0,\ \ \frac{\partial u}{\partial x}(2,y)=1$$
$$2u(x,1)+3\frac{\partial u}{\partial y}(x,1)=0,\ \ u(x,2)=\cos x.$$

Задача 2.2.

$$[A;B]\times[C;D]=[0.5;1.5]\times[0.5;1.8],$$
$$k_x=1+\frac{x}{2}+y,\ \ k_y=1+\frac{3xy}{10},\ \ q=-\frac{x+y}{2},\ \ f=x+y,$$
$$\frac{\partial u}{\partial x}(0.5;y)=0,\ \ u(1.5,y)=1$$
$$u(x;0.5)=x^2,\qquad 3u-2\frac{\partial u}{\partial y}(x,1.8)=1.$$

Задача 2.3.

$$[A;B]\times[C;D]=[1;3]\times[1;2.5],$$
$$k_x=1+e^{0.5x+y},k_y=1+\ln(1+x+y),$$
$$q=-(1+x^2+y),\ \ f=e^{-xy}$$
$$u(1,y)=\sin y,\ \ \frac{\partial u}{\partial x}(3,y)=y,$$
$$2u-\beta\frac{\partial u}{\partial y}(x,1)=\cos x\,,\ \ u(x,2.5)=1$$

Задача 2.4

$$[A;B]\times[C;D]=[0.8,2.5]\times[1,2],$$
$$k_x=1+x^2+0.2y^2,\ \ k_y=1+0.5xy,\ \ q=-(x+y),\ \ f=\cos xy$$

$$\frac{\partial u}{\partial x}(0.8,y)=0,\ \ u(2.5,y)=\sin y\,,$$

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$$u(x,1)=0,\ \ 4u-\frac{\partial u}{\partial y}(x,2)=1.$$

Задача 2.5

$$[A;B]\times[C;D]=[1,2]\times[0.7,1.8],$$
$$k_x=1+\sqrt{x+y},\ \ k_y=1+0.5(x+y),\ \ q=-xy,\ \ f=x^2+y^2$$

$$u(1,y)=y,\ \ \frac{\partial u}{\partial x}(2,y)=1,$$
$$u(x,0.7)=\cos x\,,\ \ u+2\frac{\partial u}{\partial y}(x,1.8)=0$$

Задача 2.6

$$[A;B]\times[C;D]=[0.6,1.9]\times[1,2.2],$$
$$k_x=1+0.5(x+y),\ \ k_y=1+0.3xy,\ \ q=-0.2(x^2+y^2),\ \ f=\sin(x+y)$$
$$u+3\frac{\partial u}{\partial x}(0.6,y)=0,\ \ u(1.9,y)=1$$
$$\frac{\partial u}{\partial y}(x,1)=0,\ \ u(x,2.2)=\sin x$$

Задача 2.7

$$[A;B]\times[C;D]=[1.1,2.3]\times[1,2],$$
$$k_x=1+\ln(1+x+y),\ \ k_y=1+\sqrt{x^2+y^2},\qquad q=-(x+y),\ \ f=\cos(x+y)$$

$$u(1.1,y)=0,\ \ 2u-3\frac{\partial u}{\partial x}(2.3,y)=1,$$
$$u(x,1)=x,\ \ \frac{\partial u}{\partial y}(x,2)=1$$

Задача 2.8

$$[A;B]\times[C;D]=[0.9,2]\times[0.5,2.1],$$
$$k_x=1+0.5(x^2+y^2),\ \ k_y=1+0.5xy,\ \ q=-0.1(x+y),f=e^{-x-y}$$
$$u(0.9,y)=\cos y\,,\frac{\partial u}{\partial x}(2,y)=0,$$
$$2u+\frac{\partial u}{\partial y}(x,0.5)=1,\ \ u(x,2.1)=x^2$$

Задача 2.9

$$[A;B]\times[C;D]=[1,3]\times[1,2.5],$$
$$k_x=1+\sqrt{1+x^2+y^2},\ \ k_y=1+xy+1,\ \ q=-(x+y),\ \ f=\sin xy$$

$$\frac{\partial u}{\partial x}(1,y)=1, u(3,y)=\cos y,$$
$$u(x,1)=0,\ \ 3u-2\frac{\partial u}{\partial y}(x,2.5)=0$$

Задача 2.10

$$[A;B]\times[C;D]=[1,2]\times[0.8,2],$$
$$k_x=1+x+0.5y,\ \ k_y=1+y+0.5x,\ \ q=-(x^2+y^2),\ \ f=\cos(x+y)$$

$$u(1,y)=0,\ \ \frac{\partial u}{\partial x}(2,y)=y,$$
$$u+2\frac{\partial u}{\partial y}(x,0.8)=x,\ \ u(x,2)=\sin x$$

Задача 2.11

$$[A;B]\times[C;D]=[0.7,1.6]\times[0.8,1.9],$$
$$k_x=1+0.5x+0.8y,\ \ k_y=1+0.3xy,\ \ q=-(x+y),\ \ f=\sin x+\cos y$$

$$u(0.7,y)=0,\frac{\partial u}{\partial x}(1.6,y)=y,$$
$$u+3\frac{\partial u}{\partial y}(x,0.8)=1,\ \ u(x,1.9)=\cos xy$$

Задача 2.12

$$[A;B]\times[C;D]=[1.2,2.2]\times[1.0,2.3],$$
$$k_x=1+\sqrt{x+y},\ \ k_y=1+xy,\ \ q=-(x^2+y^2),\ \ f=x^2-y^2$$

$$u(1.2,y)=\sin y,\qquad \frac{\partial u}{\partial x}(2.2,y)=0$$
$$u(x,1.0)=x,\ \ 2u+3\frac{\partial u}{\partial y}(x,2.3)=0$$

Задача 2.13

$$[A;B]\times[C;D]=[0.9,2.5]\times[0.6,1.7],$$
$$k_x=1+x^2+0.5y^2,\ \ k_y=1+0.5xy,\ \ q=-(0.5x+0.5y),\ \ f=e^{-xy}$$

$$u-\frac{\partial u}{\partial x}(0.9,y)=1, u(2.5,y)=\cos y,$$

$$\frac{\partial u}{\partial y}(x, 0.6) = 0, u(x, 1.7) = x^2$$

Задача 2.14

$$[A; B] \times [C; D] = [1.0,2.8] \times [1.0,2.0],$$
$$k_x = 1 + 0.3x + 0.4y, \;\; k_y = 1 + 0.2xy, \;\; q = -(x + y + 1), \;\; f = \sin(x + y)$$

$$u(1.0, y) = 0, \frac{\partial u}{\partial x}(2.8, y) = y\text{^}2,$$
$$u - 2\frac{\partial u}{\partial y}(x, 1.0) = \cos x\,, u(x, 2.0) = \sin xy$$

Задача 2.15

$$[A; B] \times [C; D] = [0.8,2.0] \times [0.5,1.5], k_x = 1 + \sqrt{x^2 + y^2},$$
$$k_y = 1 + 0.5xy, \;\; q = -(0.5x^2 + 0.5y^2), \;\; f = \cos x + y$$

$$u + 2\frac{\partial u}{\partial x}(0.8, y) = 0, \;\; u(2.0, y) = 1,$$
$$\frac{\partial u}{\partial y}(x, 0.5) = 0, \;\; u(x, 1.5) = \cos x$$

Завдання 3. **Рівняння Пуассона, яке задане у полярній системі координат**
Загальний вигляд рівняння:

$$\Delta u(r, \varphi) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \varphi^2} = f(r, \varphi),$$

Задача 3.1

$$r \in [1; 2], \varphi \in \left[0, \frac{\pi}{2}\right], f(r, \varphi) = \sin r \cos \varphi$$
$$u(1, \varphi) = 0, \;\; \frac{\partial u}{\partial r}(2, \varphi) = \varphi,$$
$$u - \frac{\partial u}{\partial \varphi}(r, 0) = 1, 2u + 3\frac{\partial u}{\partial \varphi}\left(r, \frac{\pi}{2}\right) = 0$$

Задача 3.2

$$r \in [0.5,1.5], \varphi \in \left[0, \frac{\pi}{2}\right], f(r, \varphi) = r^2 + \varphi^2$$
$$2u - 3\frac{\partial u}{\partial r}(0.5, \varphi) = 1, \;\; u(1.5, \varphi) = \sin \varphi,$$
$$\frac{\partial u}{\partial \varphi}(r, 0) = 0, \;\; u(r, 2\pi) = r$$

Задача 3.3

$$r \in [1;3], \varphi \in [0,\frac{\pi}{3}], f(r,\varphi) = e^{-r}\cos\varphi$$

$$u(1,\varphi) = \varphi, \quad \frac{\partial u}{\partial r}(3,\varphi) = 1$$

$$2u + \frac{\partial u}{\partial \varphi}(r,0) = 0, \quad u - 2\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{3}\right) = 2$$

Задача 3.4

$$r \in [0.7,2.0], \varphi \in \left[0;\frac{\pi}{4}\right], f(r,\varphi) = r\sin\varphi$$

$$u(0.7,\varphi) = 0, \quad \frac{\partial u}{\partial r}(2.0,\varphi) = \varphi^2,$$

$$2u + 3\frac{\partial u}{\partial \varphi}(r,0) = 1, \quad u - 2\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{4}\right) = 0,$$

Задача 3.5

$$r \in [1.2;2.5], \varphi \in \left[0,\frac{\pi}{5}\right], f(r,\varphi) = r^2\cos\varphi$$

$$u(1.2,\varphi) = \sin\varphi\,, \frac{\partial u}{\partial r}(2.5,\varphi) = 1,$$

$$u + 2\frac{\partial u}{\partial \varphi}(r,0) = 0, \quad 2u - 3\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{5}\right) = \varphi$$

Задача 3.6

$$r \in [0.6,1.8], \varphi \in \left[0,\frac{\pi}{2}\right], f(r,\varphi) = \sin(r^2+\varphi^2)$$

$$u + \frac{\partial u}{\partial r}(0.6,\varphi) = 1, \quad u(1.8,\varphi) = \cos\varphi\,,$$

$$\frac{\partial u}{\partial \varphi}(r,0) = 0, \quad u\left(r,\frac{\pi}{2}\right) = r^2$$

Задача 3.7

$$r \in [1.0,2.2], \varphi \in \left[0,\frac{\pi}{4}\right], f(r,\varphi) = r\cos\varphi + \sin r$$

$$u(1.0,\varphi) = 0, \quad \frac{\partial u}{\partial r}(2.2,\varphi) = r,$$

$$-2u + \frac{\partial u}{\partial \varphi}(r,0) = 1, \quad 2u + 3\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{4}\right) = 0$$

Задача 3.8

$$r \in [0.8,2.0], \varphi \in \left[\frac{\pi}{2}; \pi\right], f(r,\varphi) = e^{-r}\sin\varphi$$

$$u + \frac{\partial u}{\partial r}(0.8,\varphi) = 0, \;\; u(2.0,\varphi) = 1,$$
$$\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{2}\right) = 0, \;\; u(r,\pi) = r.$$

Задача 3.9

$$r \in [1.1,2.3], \varphi \in \left[0,\frac{\pi}{3}\right], f(r,\varphi) = r^2 + \sin\varphi$$
$$u(1.1,\varphi) = 0, \qquad \frac{\partial u}{\partial r}(2.3,\varphi) = \varphi,$$
$$3u + \frac{\partial u}{\partial \varphi}(r,0) = 1, \qquad -u + 2\frac{\partial u}{\partial \varphi}\left(r,\frac{\pi}{3}\right) = 0$$

Задача 3.10

$$r \in [0.9,1.9], \varphi \in \left[0,\frac{\pi}{3}\right], f(r,\varphi) = r\sin r + \cos\varphi$$

$$2u + \frac{\partial u}{\partial r}(0.9,\varphi) = 0, \;\; u(1.9,\varphi) = r,$$
$$\frac{\partial u}{\partial \varphi}(r,0) = 0, \;\; u\left(r,\frac{\pi}{3}\right) = \sin r,$$

Завдання 4. Рівняння Пуассона, яке задане у полярній системі координат (*осесиметричний випадок*)

Загальний вигляд рівняння:

$$\Delta u(r) = \frac{1}{r}\frac{d}{dr}\left(r\frac{du}{\partial r}\right) = f(r),$$

Задача 4.1

$$r \in [1,2], f(r) = r^2$$
$$u(1) = 0, \;\; \frac{du}{dr}(2) = 1.$$

Задача 4.2

$$r \in [0.5,1.5], f(r) = \sin r$$
$$2u(0.5) - 3\frac{du}{dr}(0.5) = 1, \;\; u(1.5) = 0.$$

Задача 4.3

$$r \in [0.8,2.0], f(r) = e^{-r}$$

$$u(0.8) = 1, \frac{du}{dr}(2.0) = 0$$

Задача 4.4

$$r \in [1,3], f(r) = r$$
$$u(1) + \frac{du}{dr}(1) = 0, \;\; u(3) = \cos 3$$

Задача 4.5

$$r \in [0.7,2.2], f(r) = r^2 + 1$$
$$u(0.7) = 0, \;\; \frac{du}{dr}(2.2) = 2$$

Задача 4.6

$$r \in [0.6,1.8], f(r) = \sin r^2$$
$$u(0.6) + 3\frac{du}{dr}(0.6) = 1, \;\; u(1.8) = 0$$

Задача 4.7

$$r \in \; [1.0,2.2], f(r) = \cos r$$
$$u(1.0) = 0, \;\; \frac{du}{dr}(2.2) = 1$$

Задача 4.8

$$r \in [0.9,1.9], f(r) = r^3$$
$$u(0.9) + \frac{du}{dr}(0.9) = 0, u(1.9) = 1$$

Задача 4.9

$$r \in [1.2,2.5], f(r) = r + 1$$
$$u(1.2) = 0, \frac{du}{dr}(2.5) = 2$$

Задача 4.10

$$r \in [0.8,2.0], f(r) = e^{-r^2}$$
$$-3u(0.8) + 2\frac{du}{dr}(0.8) = 1, \;\; u(2.0) = 0$$

Задача 4.11

$$r \in [1.0,2.3], f(r) = \sin r + r$$
$$u(1.0) = 0, \;\; \frac{du}{dr}(2.3) = 1$$

Задача 4.12

$$r \in [0.7,1.7], f(r) = r^2 + \sin r$$

$$-2u(0.7) + \frac{du}{dr}(0.7) = 0,\ \ u(1.7) = 0$$

Задача 4.13

$$r \in [0.6,2.0], f(r) = re^{-r}$$

$$u(0.6) = 1, \frac{du}{dr}(2.0) = 0$$

Задача 4.14

$$r \in [0.9,2.1], f(r) = r^2 - 1$$

$$u(0.9) + 2\frac{du}{dr}(0.9) = 1, u(2.1) = 0$$

Задача 4.15

$$r \in [1.1,2.4], f(r) = \cos r + 1,$$

$$u(1.1) = 0,\ \ \frac{du}{dr}(2.4) = 1$$

***Завдання* 5. Рівняння Пуассона у декартових тривимірних координатах**

Загальна форма рівняння

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f(x,y,z)$$

Задача 5.1

$$x \in [0;1], y \in [0;1], z \in [0;1], f(x,y,z) = x + y + z$$

$$u(0,y,z) = 0,\ \ \frac{\partial u}{\partial x}(1,y,z) = 1$$

$$2u(x,0,z) - 3\frac{\partial u}{\partial y}(x,0,z) = 1,\ \ u(x,1,z) - 3\frac{\partial u}{\partial y}(x,1,z) = 2$$

$$u(x,y,0) = 0,\ \ \frac{\partial u}{\partial z}(x,y,1) = 1$$

Задача 5.2

$$x \in [1;2], y \in [1;2], z \in [0.5;1.5], f(x,y,z) = x^2 + y^2 + z^2$$

$$u(1,y,z) + \frac{\partial u}{\partial x}(1,y,z) = 1,\ \ u(2,y,z) = y$$

$$\frac{\partial u}{\partial y}(x,1,z) = 0,\ \ u(x,2,z) = x$$

$$u(x,y,0.5) = 1,\ \ u(x,y,1.5) - \frac{\partial u}{\partial z}(x,y,1.5) = 2$$

Задача 5.3

$$x \in [0.5,1.5], y \in [0.5,1.5], z \in [0.5,1.5],$$

$$f(x,y,z) = \sin x \cos y + z$$

$$u(0.5,y,z)=0,\quad \frac{\partial u}{\partial x}(1.5,y,z)=1$$
$$2u(x,0.5,z)+3\frac{\partial u}{\partial y}(x,0.5,z)=1,\qquad u(x,1.5,z)-2\frac{\partial u}{\partial y}(x,1.5,z)=2$$
$$u(x,y,0.5)=0,\quad \frac{\partial u}{\partial z}(x,y,1.5)=1$$

Задача 5.4

$$x\in[0;2],y\in[0,2],z\in[0,1],$$
$$f(x,y,z)=e^{-x-y-z}$$
$$u(0,y,z)+\frac{\partial u}{\partial x}(0,y,z)=1,\ \ u(2,y,z)=1$$
$$\frac{\partial u}{\partial y}(x,0,z)=0,\ \ u(x,2,z)-2\frac{\partial u}{\partial y}(x,2,z)=1$$
$$u(x,y,0)=0,\quad \frac{\partial u}{\partial z}(x,y,1)=2$$

Задача 5.5

$$x\in[1,3],y\in[0.5,2.5],z\in[1,2],$$
$$f(x,y,z)=x+y\text{^}2+z^3$$
$$u(1,y,z)=0,\quad \frac{\partial u}{\partial x}(3,y,z)=2$$
$$u(x,0.5,z)+3\frac{\partial u}{\partial y}(x,0.5,z)=1,\ \ u(x,2.5,z)=x$$
$$4u(x,y,1)-3\frac{\partial u}{\partial z}(x,y,1)=0,\ \ u(x,y,2)=1$$

Задача 5.6

$$x\in[0.5,1.5],y\in[1,2],z\in[0.5,1.5],$$
$$f(x,y,z)=\sin xy+z$$
$$u(0.5,y,z)=0,\quad \frac{\partial u}{\partial x}(1.5,y,z)=1$$
$$u(x,1,z)+2\frac{\partial u}{\partial y}(x,1,z)=1,\ \ u(x,2,z)=0$$
$$u(x,y,0.5)=0,\quad \frac{\partial u}{\partial z}(x,y,1.5)=1$$

Задача 5.7

$$x\in[0,1],y\in[0,1],z\in[0,1],$$
$$f(x,y,z)=x^2+y^2+z^2$$

$$u(0,y,z)+\frac{\partial u}{\partial x}(0,y,z)=1,\ \ u(1,y,z)=0$$
$$\frac{\partial u}{\partial y}(x,0,z)=0,\ \ u(x,1,z)=1$$
$$u(x,y,0)+3\frac{\partial u}{\partial z}(x,y,0)=1,\ \ u(x,y,1)=0$$

Задача 5.8

$$x\in[1,2],y\in[1,3],z\in[0.5,1.5],$$
$$f(x,y,z)=e^{-x}-y+z$$
$$u(1,y,z)=0,\ \ \frac{\partial u}{\partial x}(2,y,z)=1$$
$$u(x,1,z)+2\frac{\partial u}{\partial y}(x,1,z)=1,\ \ u(x,3,z)=0$$
$$u(x,y,0.5)=0,\ \ \frac{\partial u}{\partial z}(x,y,1.5)=2$$

Задача 5.9

$$x\in[0.5,2],y\in[0.5,1.5],z\in[0,1],$$
$$f(x,y,z)=\sin x+\cos y+z^2$$
$$u(0.5,y,z)+\frac{\partial u}{\partial x}(0.5,y,z)=1,\ \ u(2,y,z)=0$$
$$\frac{\partial u}{\partial y}(x,0.5,z)=0,\ \ u(x,1.5,z)=1$$
$$u(x,y,0)-3\frac{\partial u}{\partial z}(x,y,0)=0,\ \ u(x,y,1)=0$$

Задача 5.10

$$x\in[1,3],y\in[0.5,2.5],z\in[1,2],$$
$$f(x,y,z)=x\,y\,z$$
$$u(1,y,z)=0,\ \ \frac{\partial u}{\partial x}(3,y,z)=2$$
$$u(x,0.5,z)+\frac{\partial u}{\partial y}(x,0.5,z)=1,\ \ u(x,2.5,z)=x$$
$$2u(x,y,1)+3\frac{\partial u}{\partial z}(x,y,1)=0,\ \ u(x,y,2)=1$$

Задача 5.11

$$x\in[0,1],y\in[0,1],z\in[0,1],$$
$$f(x,y,z)=x+y+z$$

$$u(0,y,z)=0,\ \ \frac{\partial u}{\partial x}(1,y,z)=1$$
$$u(x,0,z)+\frac{\partial u}{\partial y}(x,0,z)=1,\ \ u(x,1,z)+2\frac{\partial u}{\partial y}(x,1,z)=2$$
$$u(x,y,0)=0,\ \ \frac{\partial u}{\partial z}(x,y,1)=1$$

Задача 5.12

$$x\in[0.5,1.5],y\in[0.5,1.5],z\in[0.5,1.5],$$
$$f(x,y,z)=x^2+y^2+z^2$$
$$3u(0.5,y,z)+2\frac{\partial u}{\partial x}(0.5,y,z)=1,\ \ u(1.5,y,z)=0$$
$$\frac{\partial u}{\partial y}(x,0.5,z)=0,\ \ u(x,1.5,z)=1$$
$$2u(x,y,0.5)+\frac{\partial u}{\partial z}(x,y,0.5)=1,\ \ u(x,y,1.5)=0$$

Задача 5.13

$$x\in[1,2],y\in[0,1],z\in[0.5,1.5],$$
$$f(x,y,z)=\sin xy+z$$

$$u(1,y,z)=0,\ \ \frac{\partial u}{\partial x}(2,y,z)=1$$
$$u(x,0,z)+2\frac{\partial u}{\partial y}(x,0,z)=1,\ \ u(x,1,z)=0$$
$$u(x,y,0.5)=0,\ \ \frac{\partial u}{\partial z}(x,y,1.5)=1$$

Задача 5.14

$$x\in[0.5,1.5],y\in[0.5,1.5],z\in[0,1],$$
$$f(x,y,z)=e^{-x}+y+z^2$$

$$2u(0.5,y,z)+\frac{\partial u}{\partial x}(0.5,y,z)=1,\ \ u(1.5,y,z)=0$$
$$\frac{\partial u}{\partial y}(x,0.5,z)=0,\ \ u(x,1.5,z)=1$$
$$-3u(x,y,0)+2\frac{\partial u}{\partial z}(x,y,0)=0,\ \ u(x,y,1)=0$$

Задача 5.15

$$x\in[0,1],y\in[0,1],z\in[0,2],$$

$$f(x,y,z) = x^2 + \sin y + z$$

$$u(0,y,z) = 0, \quad \frac{\partial u}{\partial x}(1,y,z) = 1$$

$$-3u(x,0,z) + 2\frac{\partial u}{\partial y}(x,0,z) = 1, \quad -2u(x,1,z) + \frac{\partial u}{\partial y}(x,1,z) = 0$$

$$u(x,y,0) = 0, \quad \frac{\partial u}{\partial z}(x,y,2) = 1$$